Foundations of Quantitative Investing

Politecnico di Milano

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01

Market Participants

Who trades, and why it matters for quantitative investors

> *You'll interact with these actors and your excess return will be the result of this interaction. Knowing their purposes, incentives, time horizon and constraints will be much more beneficial to your PnL than the latest fancy model.*

The Sell Side

Facilitating trades and providing services

Dealers / Market Makers

Provide liquidity by taking the opposite side of client trades. Profit from the bid–ask spread and from uninformed trading flow. They face inventory risk from holding positions. As informed agents, they derive additional edge from observing order flow.

e.g., Citadel Securities, Optiver

Brokers

Execute orders on behalf of clients without taking risk. They route orders to exchanges and OTC dealers, anonymize clients, and handle clearing and settlement. They provide custody, margin lending, short-selling location, and research services. Compensated through commissions for all these services.

e.g., Interactive Brokers, Robinhood

Broker-Dealers

Combine both functions: they trade on their own account and on behalf of clients. This creates a tension — the dealer arm may exploit broker information (e.g., front-running). The Securities Exchange Act of 1934 regulates their conduct.

e.g., JP Morgan, Goldman Sachs

The Buy Side

Note: these categories overlap — many firms span multiple roles.

Indexers

Passive investors replicating benchmarks (MSCI, S&P, Russell). Over 37% of U.S. market cap as of 2020. Key vehicles: ETFs and index mutual funds.

e.g., BlackRock, Vanguard, State Street

Hedgers

Firms reducing financial risk from core business operations. Active mainly in derivatives markets (futures, swaps, options).

e.g., Airlines hedging fuel costs

Institutional Managers

Active managers benchmarked to indices. Over 91% trail the S&P 500 over 15 years. Constrained by tracking error limits.

e.g., Mutual funds, pension managers

Asset Allocators

Manage portfolios across asset classes (equities, bonds, commodities, alternatives). Weights are constant or slowly changing.

e.g., Endowments, sovereign wealth funds

Informed Traders

Hedge funds and prop firms pursuing absolute returns. Less regulated than institutional managers. Perform price discovery and liquidity provision.

e.g., Citadel, Two Sigma, Jane Street

Retail Investors

Trade via retail brokers. ~20% of volume (2020). Consistently unprofitable on average. Their order flow is sold to dealers (PFOF).

e.g., Individual traders on Robinhood

02

Sources of Excess Returns

Why prices are predictable — and why it's still hard to profit

Six Sources of Excess Returns

Even with accurate information, not all participants can exploit it. These frictions create opportunities.

1	**Pure Arbitrage**	Exploited by sophisticated investors, often fleeting due to technology investment in low-latency trading. Examples: Same security at different prices on different venues, ETFs vs. their constituents
2	**Heterogeneous Risk Preferences**	Investors differ in risk appetite. Some are compensated for bearing risks others avoid. The distinction between risk compensation and informational edge is often unclear. Examples: market beta, short volatility strategies.
3	**Liquidity Provision**	Certain assets are illiquid or hard to trade. Investors who can tolerate illiquidity earn a premium. Even obvious mispricings can persist when shorting is impossible or prohibitively expensive.
4	**Funding Constraints**	Distressed assets may rebound, but investors with limited capital or margin constraints cannot exploit the opportunity. Deleveraging spirals amplify this effect.
5	**Flow Predictability**	Institutional investors and index funds trade known securities on known dates (e.g., index reconstitutions). Speculators provide liquidity in advance. Not front-running — based on public information.
6	**Informational Advantage**	Some investors have better, more accurate forecasts of future prices. This is the core of statistical arbitrage — predicting returns based on insights competitors lack.

03

Stylized Facts of Financial Returns

Empirical properties that any risk model must account for

Key Properties of Asset Returns

Geometric diffusion (GBM) predicts Gaussian, independent returns — reality is starkly different.

Absence of Return Autocorrelations

ACF ≈ 0

at daily freq

Lagged autocorrelations of returns are negligibly small at daily and lower frequencies. However, autocorrelation increases at time scales where market microstructure becomes relevant, i.e. intraday. Note that absence of autocorrelation doesn't imply independence!

Volatility Clustering

ACF > 0

for $|r|$ and r^2

Absolute returns $|r(t)|$ and squared returns $r^2(t)$ show strong positive autocorrelation. Large realized return movements persist over time, as do small ones. This motivates GARCH-type models.

Heavy Tails

$\alpha \approx 4$

tail index

The unconditional distribution of returns exhibits power-law tails: $P(R > x) \sim Cx^{-\alpha}$, with tail index $\alpha \approx 4$. The probability of extreme moves far exceeds what any Gaussian model predicts.

Aggregational Gaussianity

$\to N(\mu, \sigma^2)$

as horizon ↑

At longer horizons (weekly, monthly), the distribution of log-returns converges toward a Gaussian distribution. Returns become closer to normal as the observation frequency decreases.

04

Structure of a Quantitative Trading System

The building blocks inside the "black box"

> *Not all the blocks we'll review are necessary and some may be combined, but the division is still helpful because in structured organizations the processes are developed by different teams.*

The Black Box Revealed

DATA

Prices, volumes, characteristics, time series, unstructured data

Powers the entire black box machinery

Alpha Model

Predicts the future returns of the instruments the quant wants to consider trading.

Risk Model

Limits exposure to factors unlikely to generate returns but that could drive losses.

Transaction Cost Model

Estimates cost of migrating to target portfolio. Costs take different forms: bid/ask spread, slippage, and market impact.

Portfolio Construction

Balances profits, risk limits, and transaction costs to determine the optimal portfolio

Execution

Low-cost, efficient trade implementation

05

Understanding Returns of Major Asset Classes

Decomposing equity, bond, credit, and currency returns

Asset Class Return Drivers

Equities

$$R \approx \text{Dividend Yield} + \text{Dividend Growth} + \text{Valuation Change}$$

Hist: ~11%

Equity premium over cash: ~7.5%

~2% Div. Yield (today)

Dividends relative to price paid. Historically averaged 3.9% (U.S. 1926–2013). Includes net share buybacks.

~3.5% Div. Growth (nominal)

Driven by inflation (~2%) plus real economic growth (~1.5%). Historically 4.6% due to higher past inflation.

2.4% Valuation Change (hist.)

Changes in price-dividend ratio. Cannot grow forever. Mean-reversion likely reduces future contribution.

Bonds

$$R \approx YTM - Dur \times \Delta YTM$$

Long-run return equals the yield to maturity. Short-run returns depend on yield changes, driven by surprises in growth, inflation, or monetary policy. The roll-down effect assumes an unchanged yield curve.

Credit

$$R \approx Spread - Dur \times \Delta S - Loss$$

Credit return equals the spread, minus duration-weighted spread changes, minus expected default losses. A higher credit spread leads to a higher return, everything else being equal. Spread narrowing produces capital appreciation; widening causes losses.

Currency

$$E(R) \approx r^* - r - \Delta\pi + PPP$$

Short-run return equals carry (foreign interest rate) plus currency appreciation. The carry trade exploits the failure of uncovered interest rate parity. Long-run return depends on the real interest rate differential and convergence to purchasing power parity.

Key Takeaways

1. Markets are populated by diverse participants with different objectives, constraints, and information — these differences create trading opportunities.
2. Excess returns arise from arbitrage, risk premia, liquidity provision, funding constraints, flow predictability, and informational advantages.
3. Asset returns exhibit heavy tails, volatility clustering, and weak autocorrelation — simple Gaussian models are inadequate.
4. A quant trading system integrates alpha, risk, and cost models into a portfolio construction framework, all powered by data.
5. Each asset class has distinct return drivers: dividends and growth for equities, yield and duration for bonds, spreads and defaults for credit, carry and PPP for currencies.

Sources

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